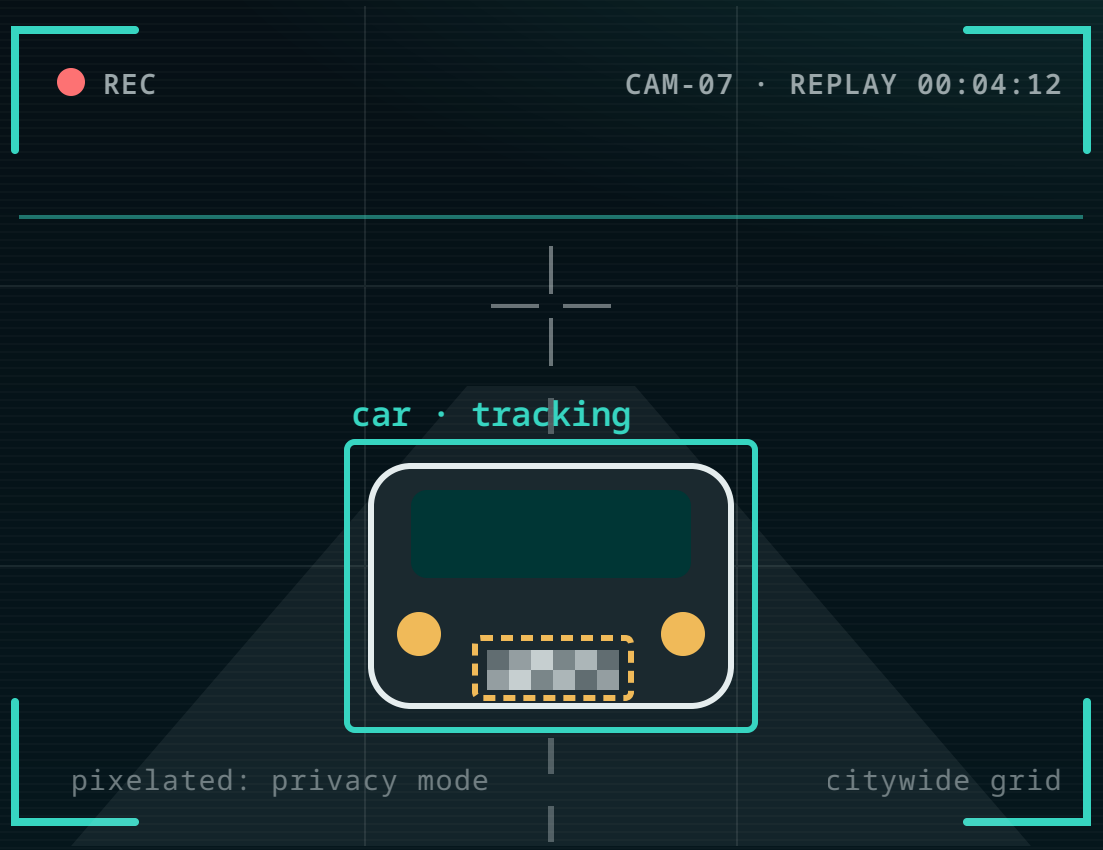
CSE311 · ARTIFICIAL INTELLIGENCE

SAKSHI

साक्षी · *witness*

Citywide vehicle analytics that records what it saw, and admits what it could not read.

- citywide camera grid · one global frame budget
- YOLOv8n · YOLOv9-tiny · MobileViT-v2
- consensus gate armed: 3 of last 8, else abstain



A S Sri Dathan 2024BCS0091
Akshay V.S 2024BCD0015
Bharadwaj B Chand 2024BCS0059
Shivansh Neekhra 2024BCS0023
Srimoneyshankar Ajith 2024BCS0055

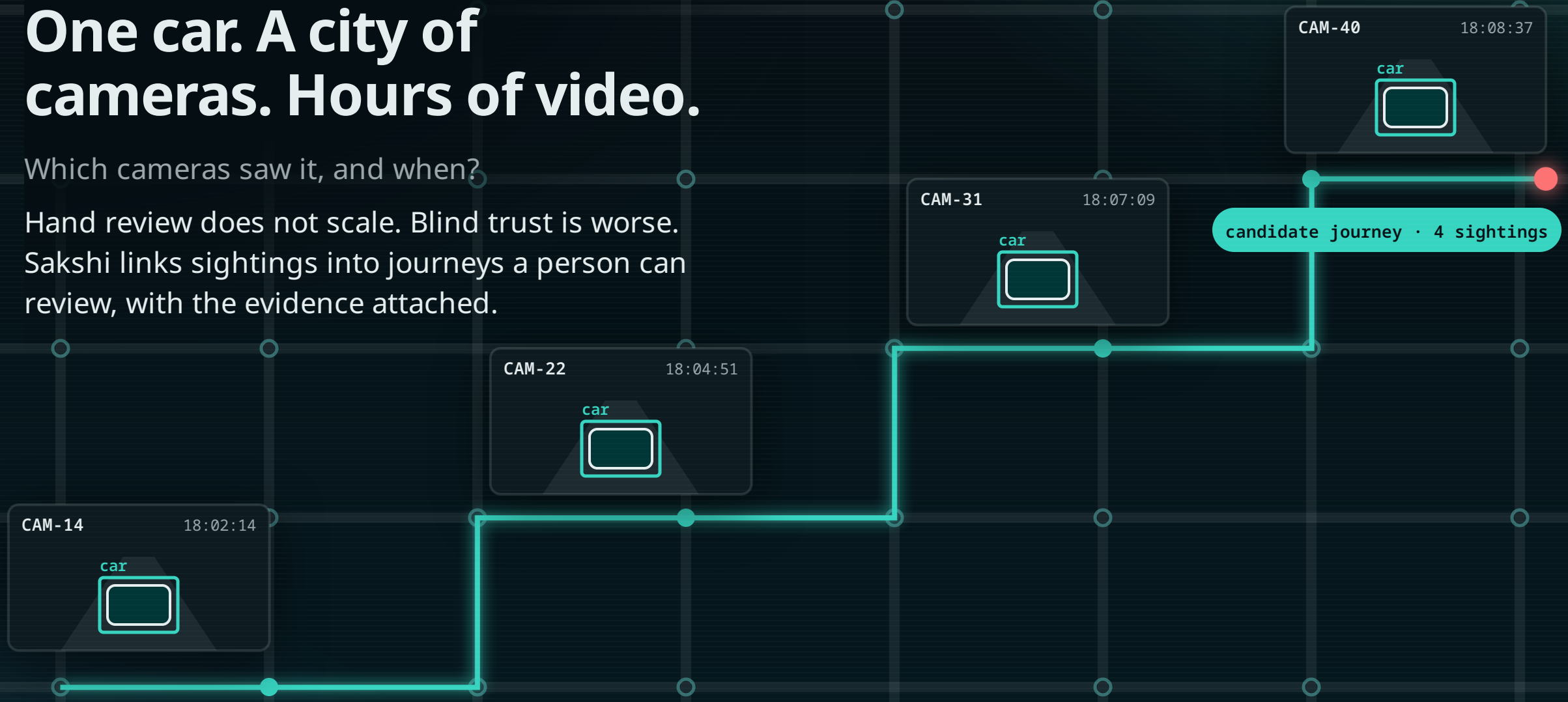
Guide: Dr. Sreeja M U · Department of CSE, IIIT Kottayam · Third year, 2026–2027

02 THE PROBLEM ILLUSTRATIVE

One car. A city of cameras. Hours of video.

Which cameras saw it, and when?

Hand review does not scale. Blind trust is worse. Sakshi links sightings into journeys a person can review, with the evidence attached.



03 INSIDE ONE CAMERA

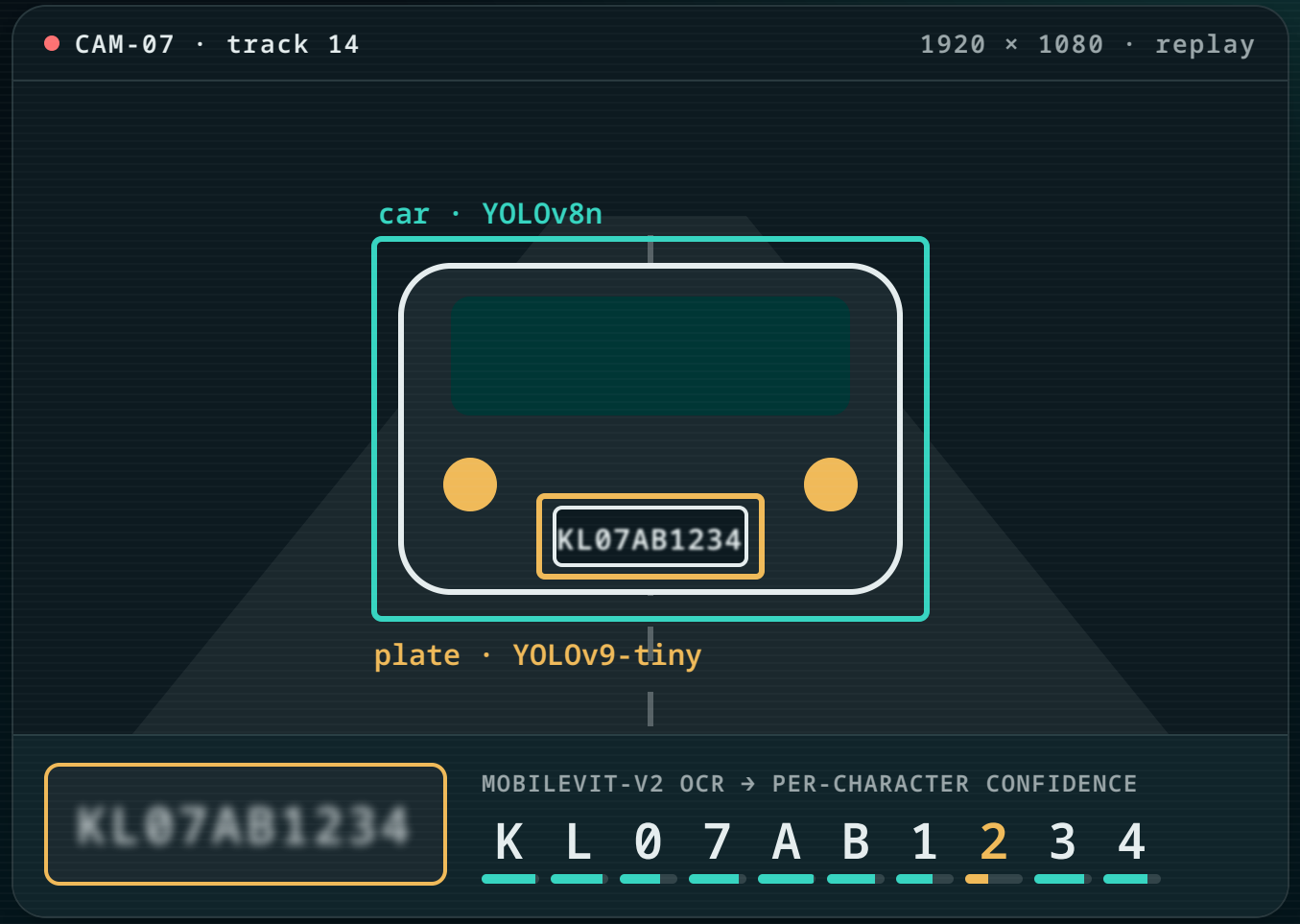
SYNTHETIC PLATE

See the car. Find the plate. Read it.

01 Detect
YOLOv8n, 640 px input. Cars, motorcycles, buses, trucks, and people for privacy masking.

02 Locate
YOLOv9-tiny plate detector at 384 × 384.

03 Read
MobileViT-v2 OCR via ONNX Runtime. Characters plus confidences.



04 PER-TRACK CONSENSUS

SYNTHETIC PLATES AND SCORES

Three reads must agree. Otherwise, abstain.

CAM-07 · track 14

#1	KL07AB1234	0.61
#2	KL07AB1Z34	0.44
#3	KL01AB1234	0.52

3 needed

KL07AB1234
KL07AB1Z34
KL01AB1234

? ABSTAIN · READS CONFLICT

CAM-08 · track 3

#1	KL07CD5678	0.58
#2	~~KL07CD5678~~	0.29 < 0.35
#3	KL07CD5678	0.47
#4	KL07CD5678	0.66

3 needed

KL07CD5678

SUPPORTED · 3 OF LAST 8

05 THE RULE EVERYTHING RESTS ON

A plate becomes an identity only when every gate agrees

read

1
Confidence
OCR score at least 0.35.

✓ ≥ 0.35

2
Format
Normalised to Indian registration syntax. Raw text and every edit kept.

✓ valid syntax

3
Clean track
Reads from an ambiguous track never count.

✓ not ambiguous

4
Consensus
3 consistent reads among the last 8, same camera, same track.

3 of 8

✕ Fail any gate: the read stays in history, raw text intact, and is excluded from plate queries.

Conflicts stay conflicts. Character suggestions are review-only.

06 INTRODUCTION

Sakshi turns multi-camera footage into reviewable, auditable vehicle observations.

01

See

Pretrained YOLOv8n finds cars, motorcycles, buses and trucks. People too, for masking.

02

Read

A plate detector and a MobileViT-v2 OCR propose text. Doubt means abstain.

03

Follow

A camera-local tracker links boxes over time by motion and geometry.

04

Witness

Every read, model hash, review and export is stored and auditable.

HOW TO READ THIS DECK MEASURED recorded on our hardware PROJECTED estimate, not tested ILLUSTRATIVE synthetic, for explanation

07 PROBLEM STATEMENT

Too much video. Too little trust in what a model says about it.

- Searching every recording by hand is slow and hard to repeat.
- OCR misreads small, blurred plates.
- A replay timestamp is not when the scene happened.
- A wrong identity costs more than no identity.

07 OBJECTIVES

SHARE	One processor samples every camera under a global frame budget.
SEE	Pretrained vehicle detection and plate OCR.
DOUBT	Keep raw and ambiguous reads. Abstain on conflict.
FOLLOW	Camera-local tracking that retires ambiguous tracks.
PROVE	Hashes, audit records, verifiable exports.
REVIEW	A dashboard with queries, rules and reports.
EVALUATE	Reproducible tools that say what they cannot measure.

08 LITERATURE

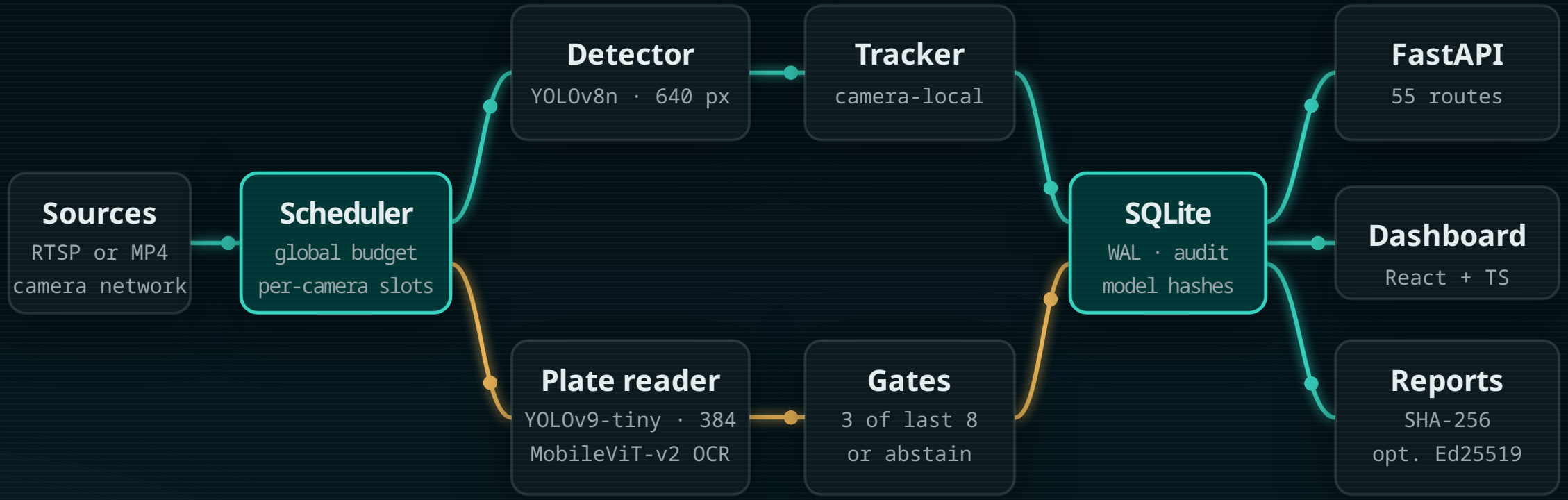
Standing on established methods

WORK	IDEA	ROLE IN SAKSHI
YOLO Redmon et al., 2016	Single-stage real-time detection	Pretrained YOLOv8n finds vehicles and people
SORT Bewley et al., 2016	Motion + global assignment	Our tracker: assignment with custom motion, geometry, ambiguity rules
DeepSORT Wojke et al., 2017	Adds learned appearance	Comparison only. No appearance model, no re-ID
YOLO ALPR Laroca et al., 2018	Detect plate, then read characters	Our two-stage plate detector + OCR
MobileViT v2 Mehta & Rastegari, 2022/23	Separable self-attention, linear cost	The plate OCR backbone
INT8 Jacob et al., 2018	Integer-only inference	Our FP32 vs INT8 detector benchmark

Gap we address: a review workflow around uncertain predictions, not a new detector.

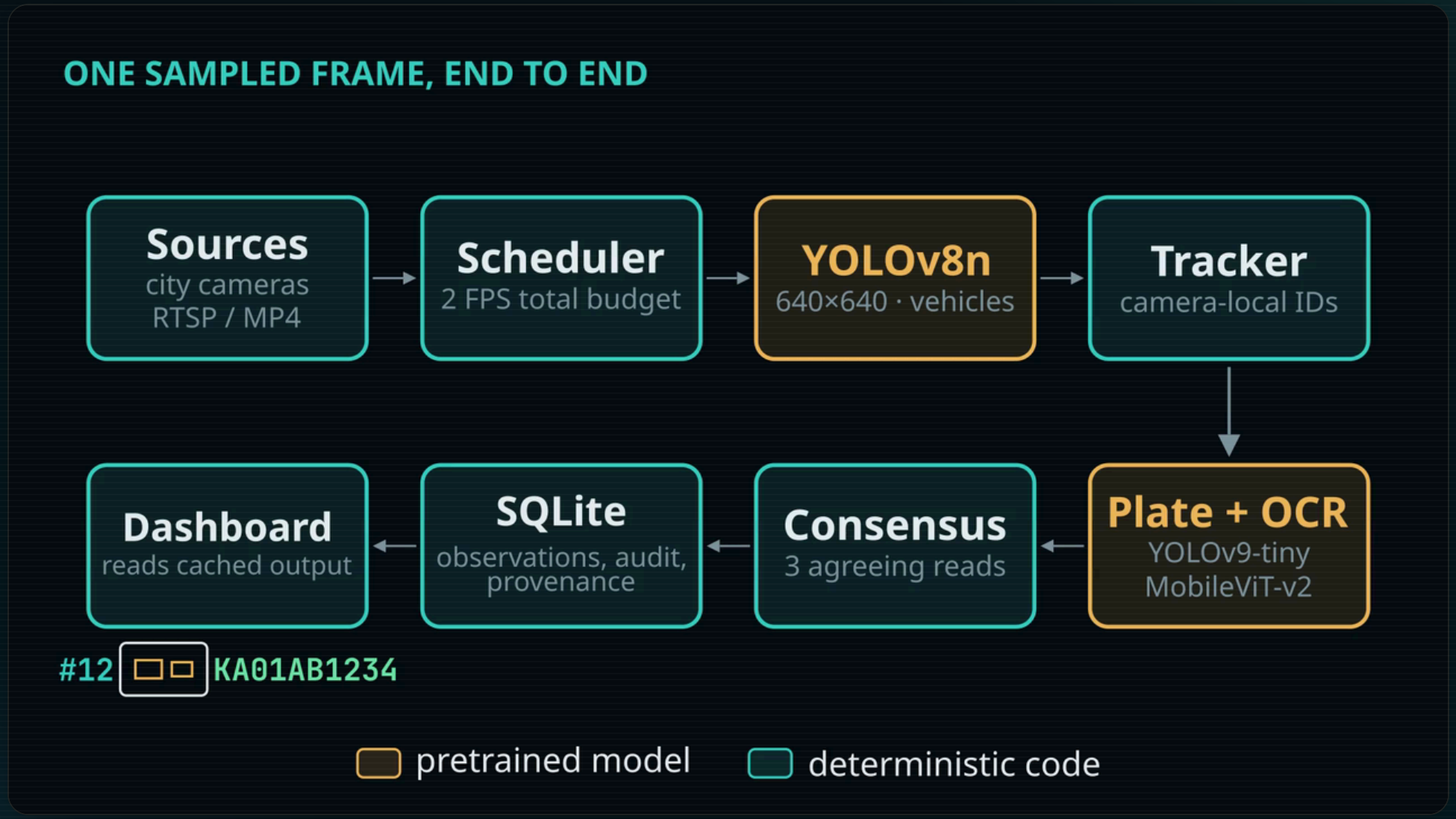
09 SYSTEM ARCHITECTURE

Inference is owned once. Everything else reads its output.



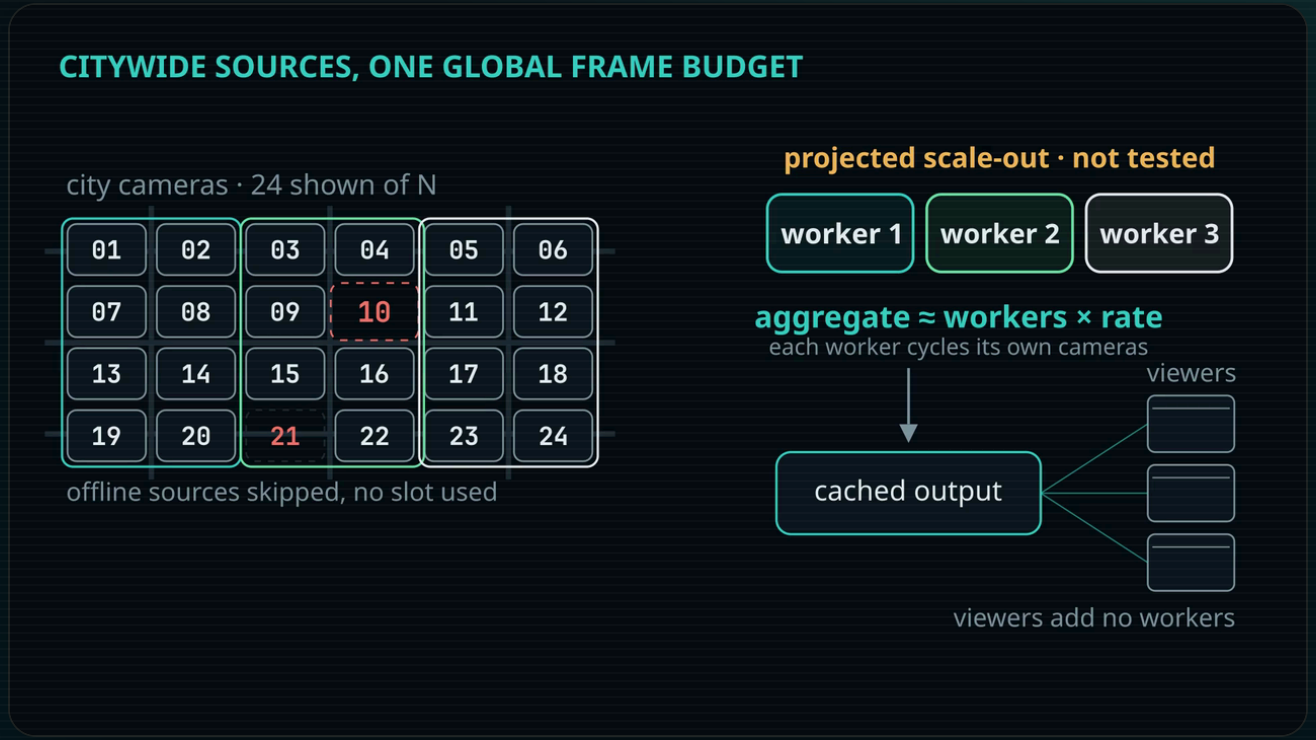
Browsers read cached frames and results. More viewers add no inference workers.

10 THE PIPELINE, ANIMATED ANIMATION



Every camera shares one frame budget

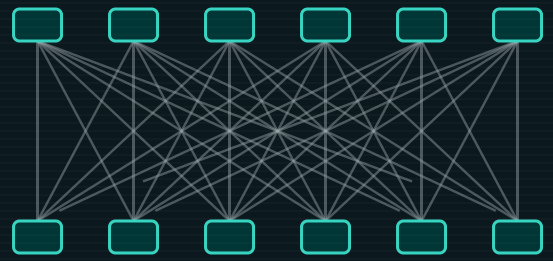
- A configurable **global** budget (default 2 frames per second), not per camera.
- Round-robin over available sources, so no camera starves.
- Missing or retrying sources do not burn inference slots.
- To grow: partition cameras across workers. Viewers read the cache and add no load.



12 AI TECHNIQUE ARCHITECTURE SKETCH

Transformer-style OCR, light enough to scale

Plate crop input › **Conv stem** local edges › **MV2 blocks** local features › **MobileViT-v2** global layout › **Char head** per position › `KL07AB1234` + confidences



Self-attention
every patch with every patch: cost grows with k^2



Separable
patches meet one context vector: cost grows with k

Pretrained (FastPlateOCR global model). We did not train it. Its accuracy on Indian plates is **unmeasured**.

13 TRACKING

Follow a car within one camera. Let go when unsure.

```
cost = 0.75·dist + 0.20·(1 − IoU) + 0.05·Δarea
```

0.75 predicted centre distance

0.20 box overlap, 1 − IoU

0.05 change in box area

Heuristic scores from `backend/tracking.py`, not identity probabilities.

CAMERA-LOCAL TRACKING



previous tracks (dashed) · new detections (solid)

```
cost =
  0.75 · predicted-centre distance
+ 0.20 · (1 − IoU)
+ 0.05 · area change
```

class, size and distance gates reject pairs

	#3	#4	new
D1	0.583	0.596	1.1

`|Δ| = 0.013 ≤ margin 0.04`

near-tie → don't guess

#3 and #4 retired · detection starts a new ID

14 EVIDENCE

HASHES SHOWN ARE ILLUSTRATIVE

Every result can say where it came from

01

Observation

Source, session, frame position, model and component digests.

sha256 **a41f09c2…**

02

Frame + crop

SHA-256-referenced JPEGs. A failed capture is recorded as missing.

sha256 **7c0e5b18…**

03

Report

Immutable snapshot, verified ZIP, expiry. Empty reports refused.

sha256 **e93d2a70…**

04

Signature

Optional Ed25519 against a trusted key. Hash-chained audit.

sha256 **15bf8d4e…**

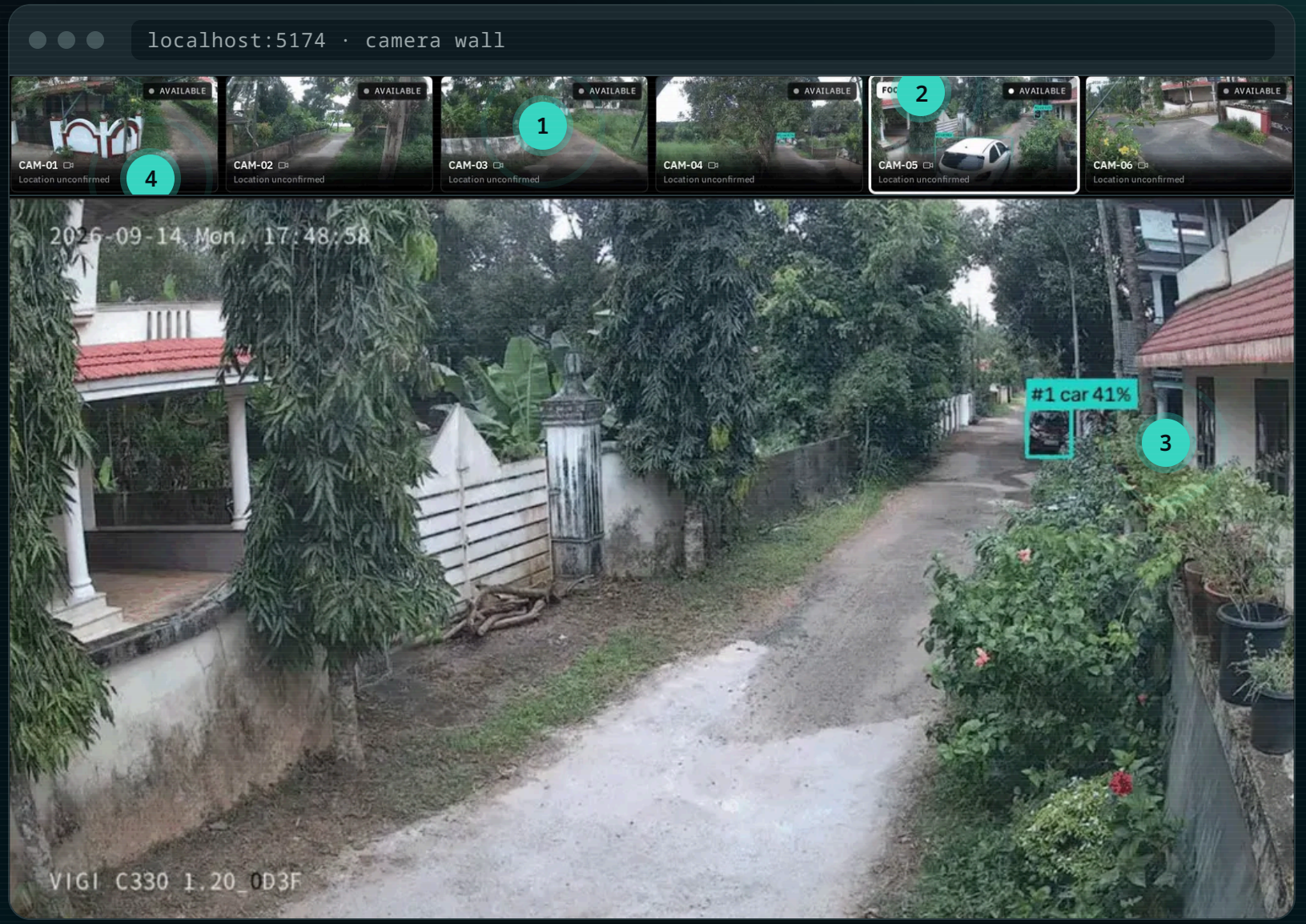
A hash detects a change. It does not prove the scene, or the plate text, is true.

15 THE APP

REAL SCREENSHOT

The camera wall

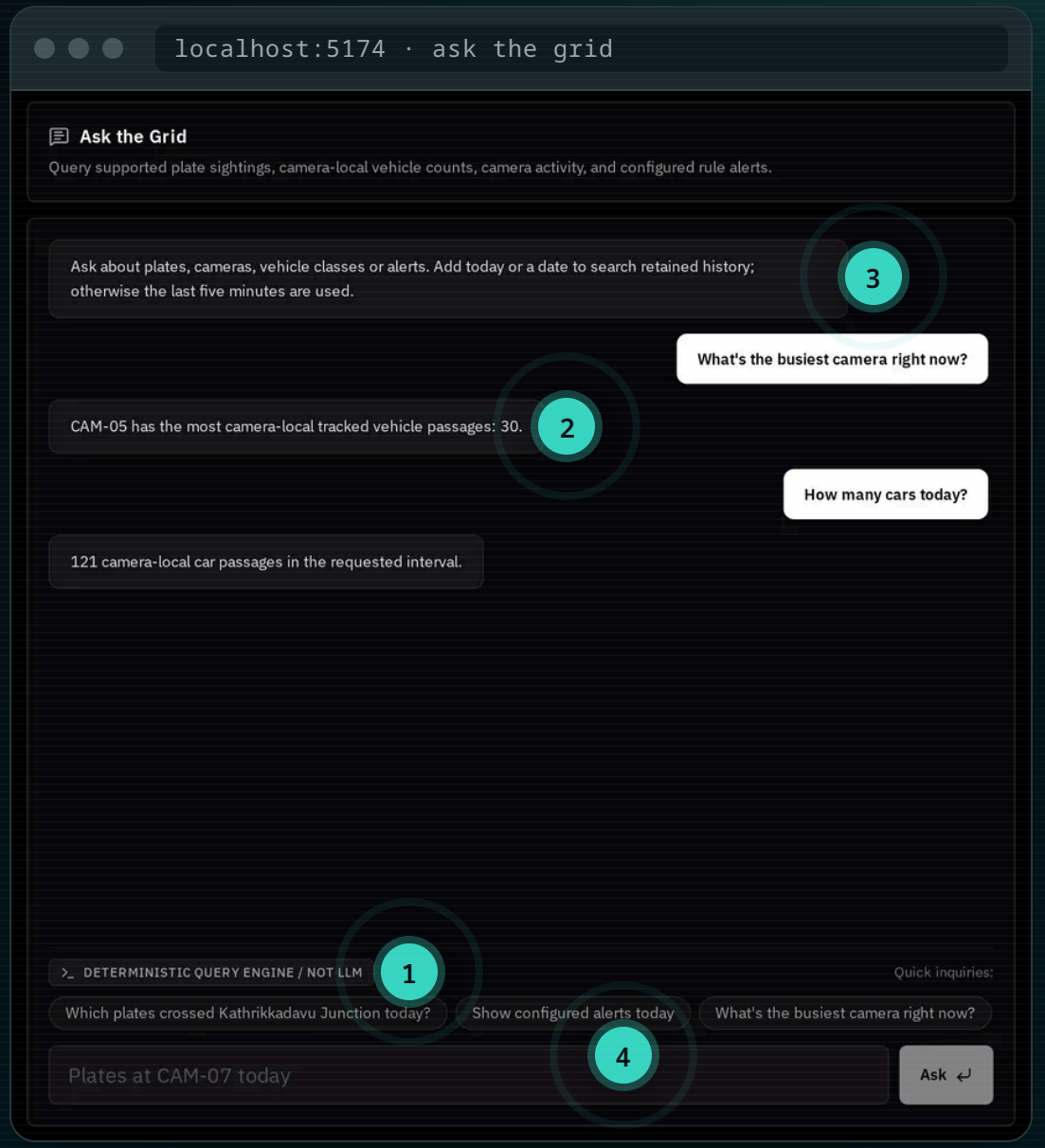
1. Every live feed on one wall, sampled by one shared processor.
2. Focus any camera to inspect its sampled detections.
3. Detections carry a track ID. Scores are model confidence, never called accuracy.
4. Unverified metadata is labelled, not hidden.



16 ASK THE GRID REAL SCREENSHOT

Questions, not prompts

1. A bounded grammar. No LLM, no generated SQL.
2. Answers say “camera-local passages”, never “unique vehicles”.
3. Explicit time window: today, a date, or the last five minutes.
4. Quick inquiries are fixed templates, not free-form prompts.

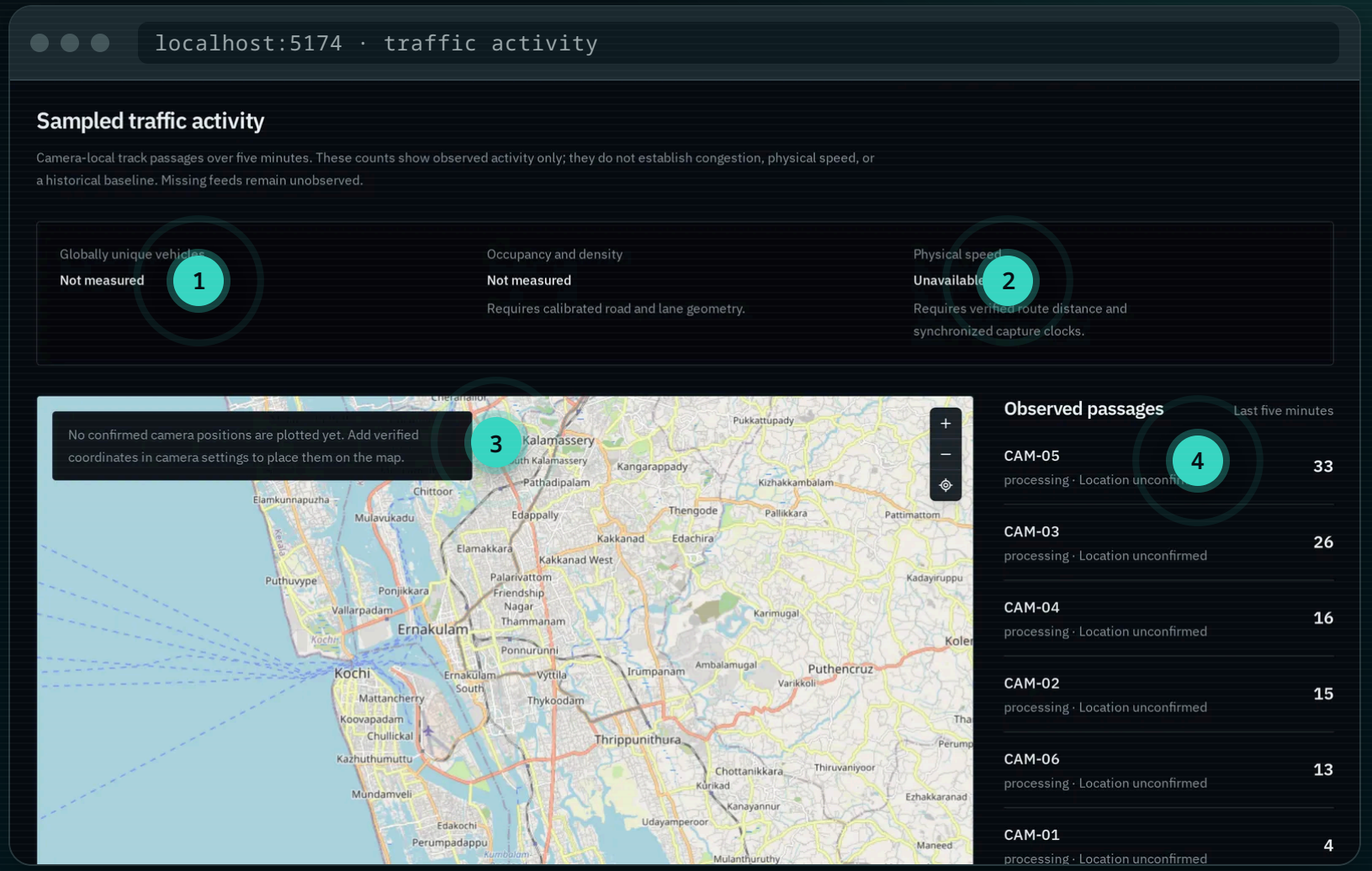


17 HONEST UI

REAL SCREENSHOT

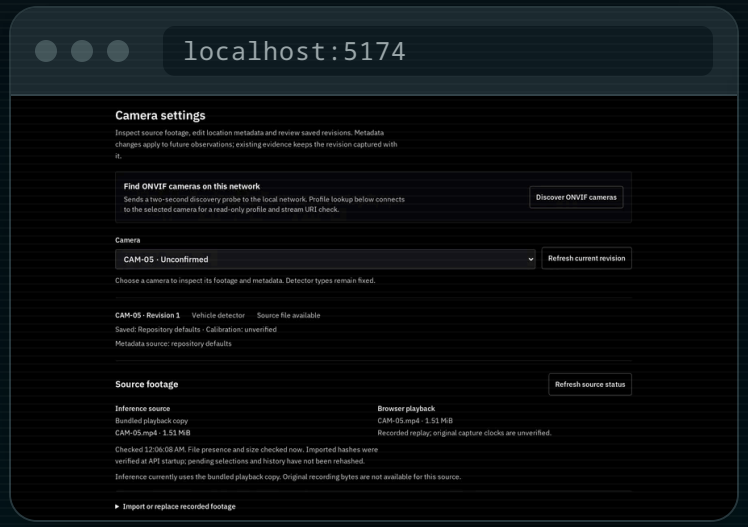
The product says “not measured”

1. Globally unique vehicles: not measured.
2. Speed needs verified distances and synced clocks. Unavailable.
3. Unverified camera positions are not plotted.
4. Counts are camera-local passages, last five minutes.



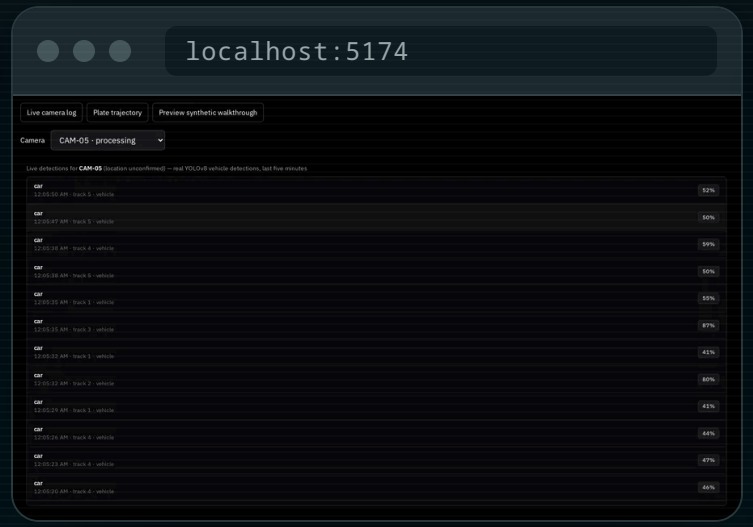
18 AND THE REST OF THE APP REAL SCREENSHOTS

Settings, timelines and alerts



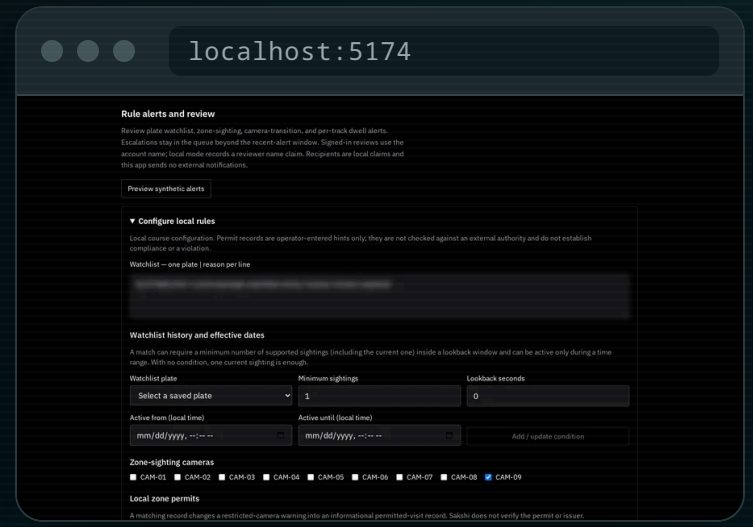
Camera settings

Audited metadata revisions. Calibration marked unverified.



Evidence timeline

Real YOLOv8 detections for one camera, track-local.



Rule alerts

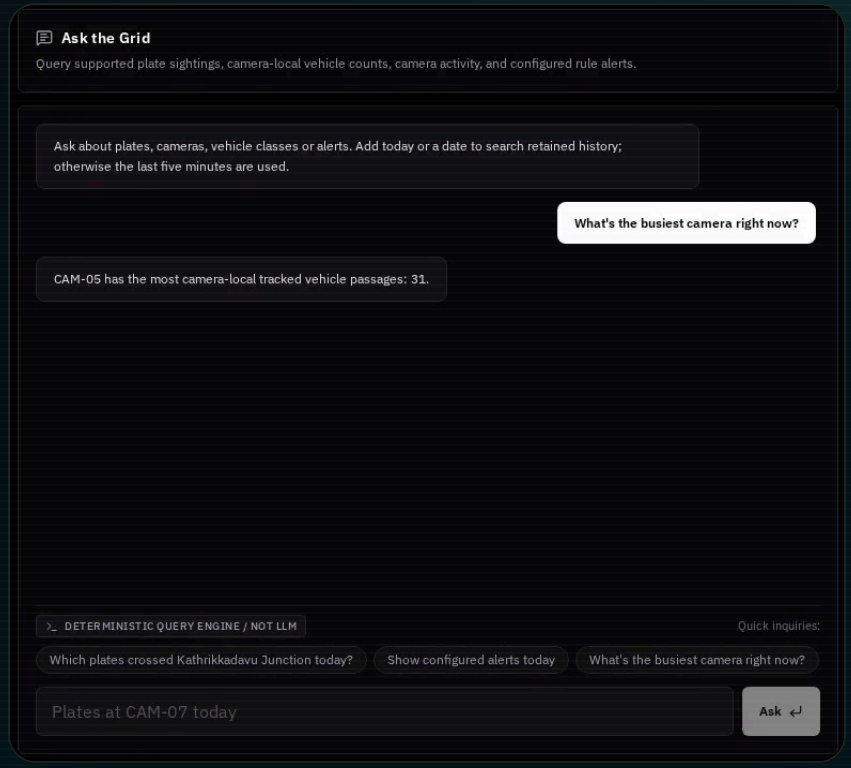
Watchlists, zones, dwell rules, a durable escalation queue.

19 THE APP, RUNNING

SCREEN RECORDING



Sampled detections updating, track IDs attached



A bounded query, answered

20 ENGINEERING AT A GLANCE REPO COUNTS

Small models. A serious system around them.

55
API routes, FastAPI

442
tests

47
test files

6
pipeline stages

3
pretrained models

0
LLMs in the query path

`Python 3.12` `PyTorch 2.14` `Ultralytics 8.4.155` `ONNX Runtime 1.30.0` `Fast-ALPR 0.4.0` `FastPlateOCR 1.1.0`

`OpenCV` `FastAPI` `SQLite WAL` `React + TypeScript` `Vite` `Docker Compose`

21 PROTOTYPE RUN · AVAILABLE SAMPLE FEEDS · FIVE MINUTES MEASURED

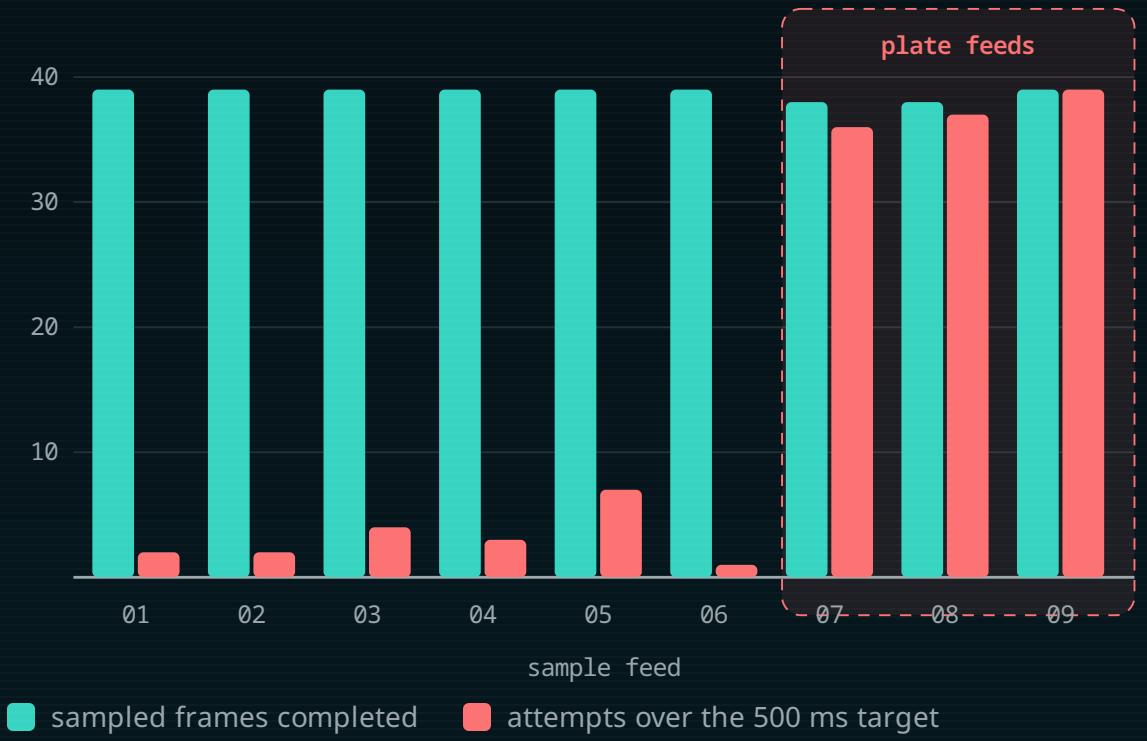
Both AI branches on, every feed processed, zero failures

349
sampled frames processed

298.011 s
valid replay window

0
failed inference attempts

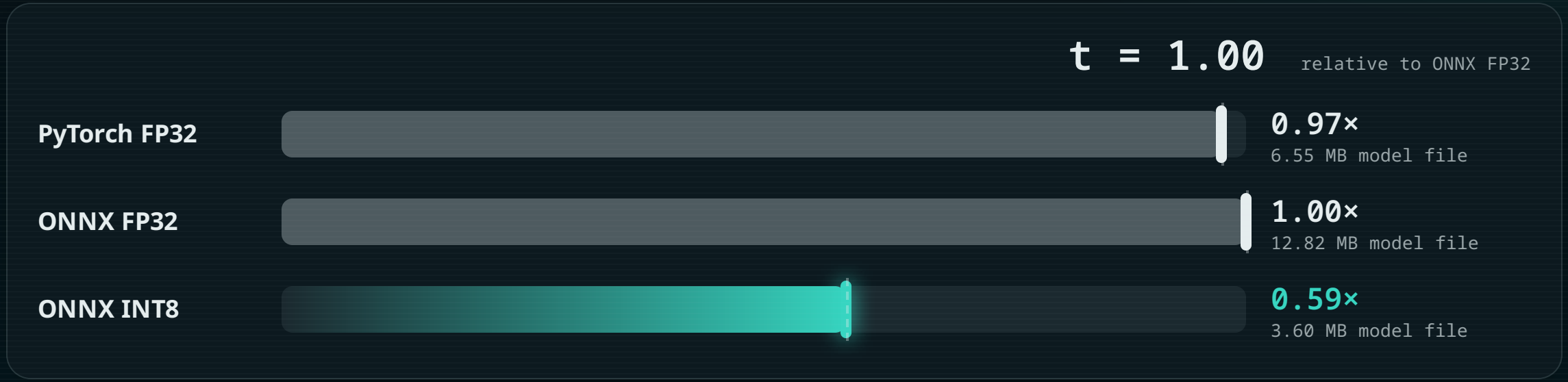
38–39
outputs per feed: balanced



Overruns cluster on full-resolution plate feeds: plate inference is the first stage to accelerate.

22 VEHICLE DETECTOR · ONE FORWARD PASS · MEDIAN OF 10 RUNS MEASURED

FP32 vs INT8: one forward pass, raced



≈ 41%
faster forward pass than ONNX FP32

12.82 → 3.60 MB
exported model file

15 of 16
detections agree with FP32. Agreement, not accuracy.

Next: go live
INT8 is benchmarked; wiring it into the pipeline is next.

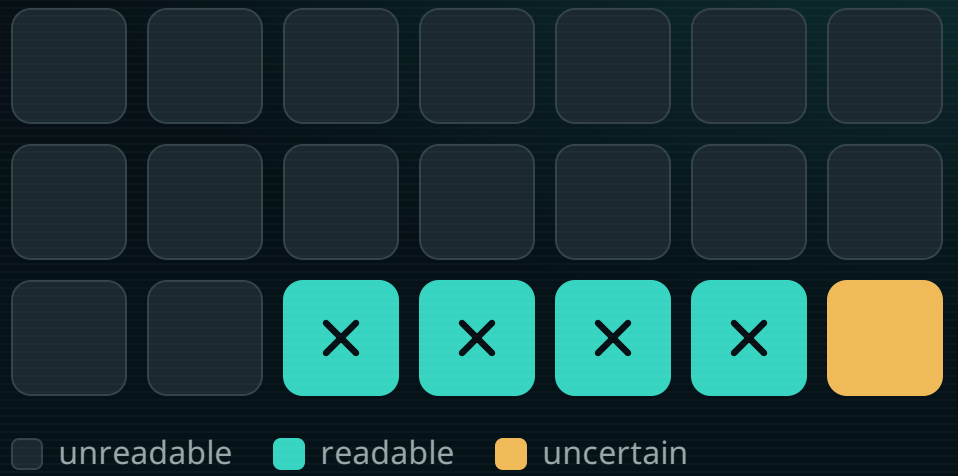
23 SELECTED CROPS · NOT REPRESENTATIVE

MEASURED

When the OCR was wrong, the gates said no

- 21 crops from CAM-07 to CAM-09: 16 unreadable, 4 readable, 1 uncertain.
- All 4 readable plates were misread by the OCR.
- **All 4 were rejected. None became an identity.**

Offline probe: 68 crop variants, 1 exact match, 5 passed format and confidence checks. Diagnostic only, not deployed.



4 of 4 mismatches rejected by the gates

Shows abstention on these crops. Not an accuracy or false-accept rate.

24 HONEST SCOPE

Built and measured today. Earned next.

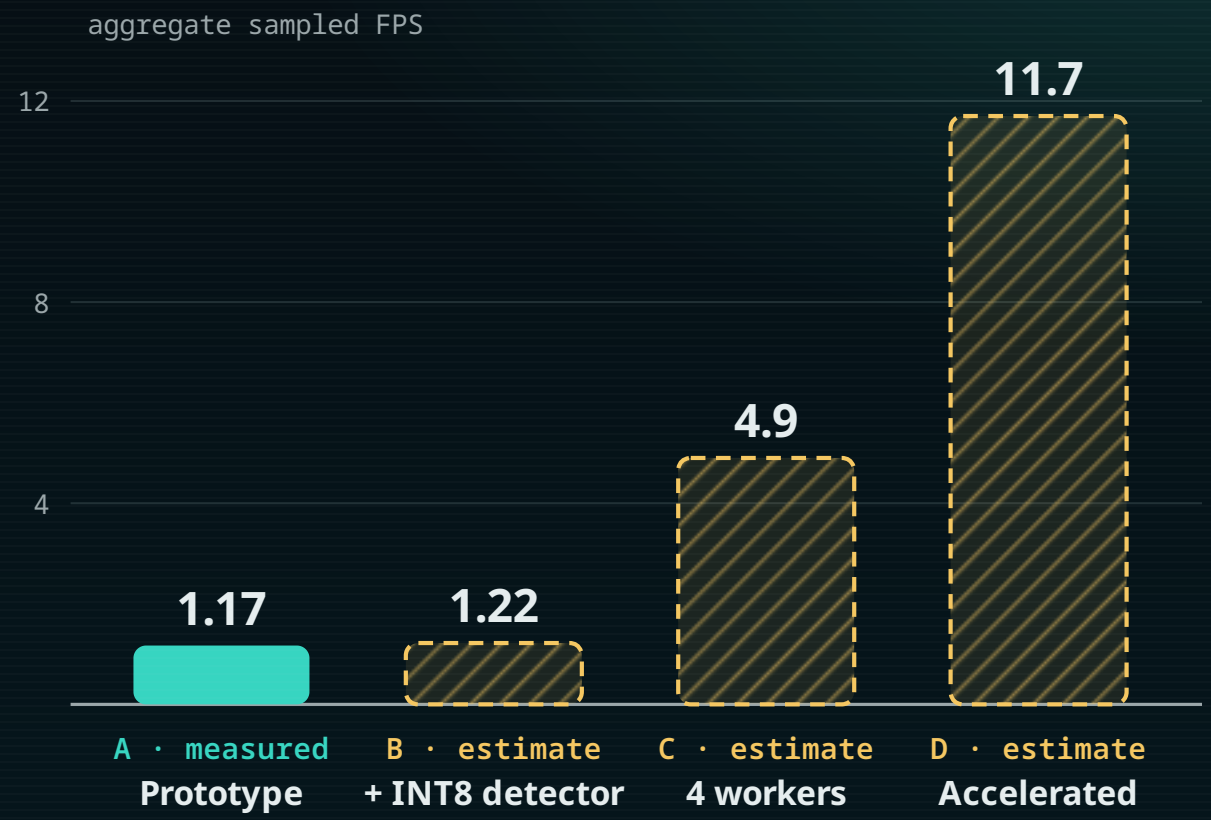
TODAY	NEXT, BEFORE WE CLAIM IT
End-to-end pipeline ran on the sample feeds with zero failed attempts.	Labelled, held-out accuracy evaluation for detection, plates and tracking.
The gates abstained on every mismatched plate read we reviewed.	Synchronised clocks and verified locations before identity or journey claims.
INT8 detector export runs about 41% faster than FP32.	INT8 and accelerated plate inference in the live pipeline.
Every observation and report is hashed, versioned and auditable.	Scale-out deployment: workers by camera, networked database.

25 SCALE OUT BY CAMERA

PROJECTED

The prototype is a starting point, not a ceiling

- **A** prototype throughput, measured.
- **B** swap in the INT8 detector.
- **C** partition cameras over 4 workers.
- **D** accelerated inference, assumed 10×.
- Decode, inference and a networked database scale independently.



26 RECAP: CONSENSUS, ANIMATED

ANIMATION

PLATE CONSENSUS · PER CAMERA, PER TRACK

OCR read

conflict → abstain
no plate asserted · reads kept for review

window · last 8 reads

KA01AB1234	0.81
KA01AB1234	0.74
KA01AB1234	0.66
KA01AB1284	0.58

gates

- confidence ≥ 0.35
- format valid
- track not ambiguous

history only
not queryable

KA01AB7234
0.22

Labels first. Then scale. Then claims.

Now
Working prototype, exercised on the available sample feeds.

Label
Authorised, prediction-blind held-out set, split by vehicle, camera, place, session.

Measure
Precision, recall, exact-plate match, with abstention coverage and intervals.

Adapt
Fine-tune OCR on licensed regional plates. Promote only after paired evaluation.

Scale out
Workers partitioned by camera, accelerated inference, networked database.

Connect
Synchronised clocks and verified locations unlock citywide journeys.

Deploy
Privacy, legal, retention and security review, then a city pilot.

28 WHAT IT BECOMES VISION

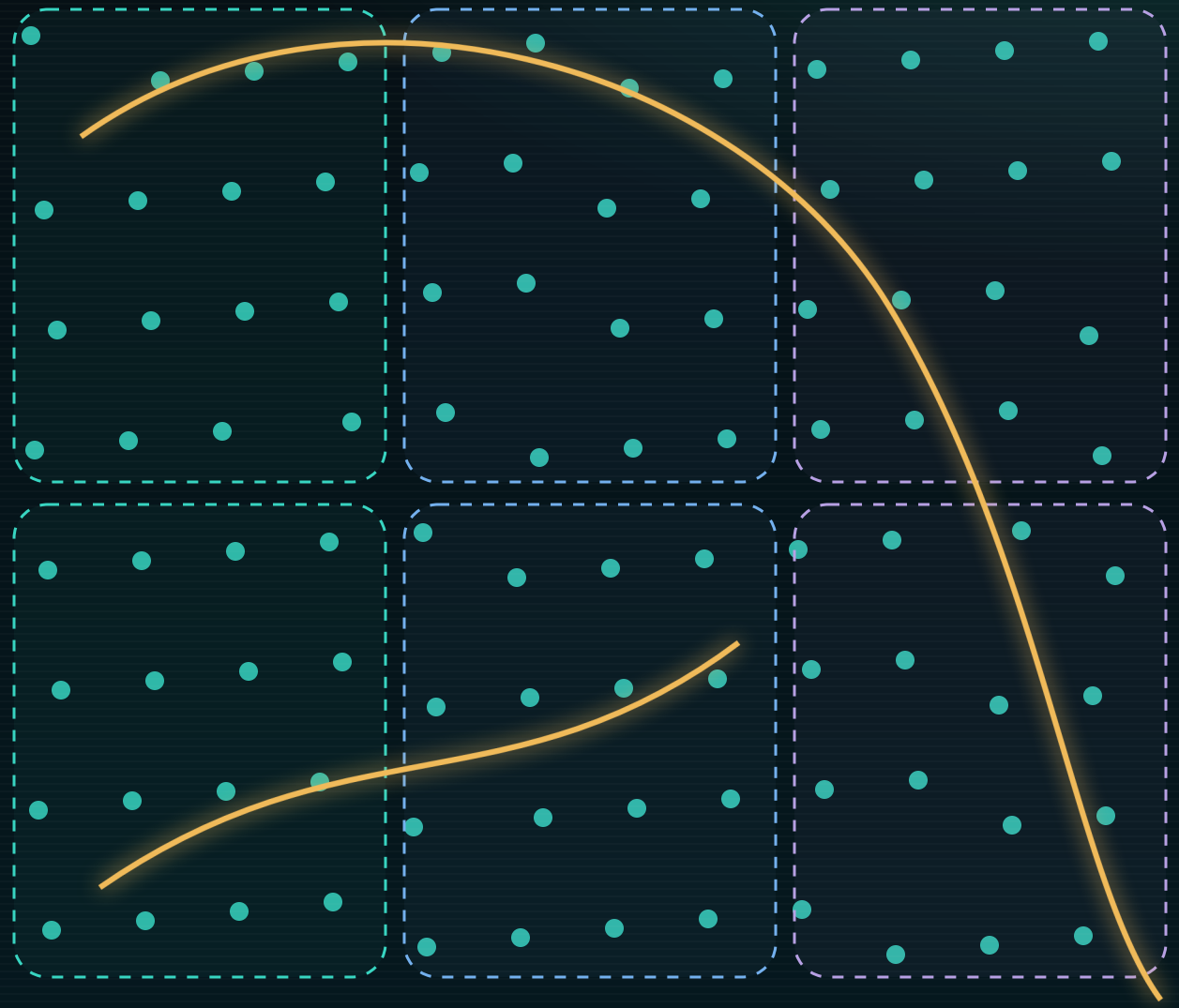
A citywide witness

Partitioned
Camera groups map to workers; each owns its tracks.

Accelerated
Plate inference moves to accelerators where resolution demands.

Connected
Journeys across the city, once clocks and locations are verified.

Accountable
One networked evidence store. Humans decide.



29 FAQ

Questions we expect

How accurate is the plate OCR?

Not yet measured, and we say so. No held-out labels exist. The 4 readable crops we checked were all misread, and all 4 were rejected.

Did you train the models?

No. YOLOv8n, YOLOv9-tiny and MobileViT-v2 are pretrained. Our work is the pipeline that keeps their doubt visible.

How does it reach city scale?

Scale out by camera: the scheduler hands camera partitions to workers, and decode, inference and storage scale independently.

Can it follow a car across the city?

That is the design. Sightings link into candidate journeys today; identity claims wait for synchronised clocks and verified locations.

Why not an AI chatbot for search?

A bounded grammar is predictable and auditable. Unsupported questions are rejected, not guessed.

Is any plate in this deck real?

No. Story plates are synthetic. The report reproduces no plate strings or private frames.

30 SAKSHI · WITNESS

It names a plate only when the evidence agrees.

Everything else stays a question for a human.

Questions?

github.com/moneytosms/ai26 · Story site · Slides PDF · Slides PPTX · Run it: uv run run.py

A S Sri Dathan · Akshay V.S · Bharadwaj B Chand · Shivansh Neekhra · Srimoneyshankar Ajith · Guide: Dr. Sreeja M U · IIIT Kottayam